

Total No. of Questions : 8]

PD4703

SEAT No. :

[Total No. of Pages : 2

[6404]-209

B.E. (Mechanical)

INDUSTRIAL ENGINEERING

(2019 Pattern) (Semester - VII) (Elective - III) (402044 D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answers Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of electronic calculator is allowed.*
- 4) *Assume suitable data, if necessary.*

Q1) a) What is the relationship between Plant Layout and Material Handling equipment. **[7]**

b) Explain the selection criterion for material handling. Explain in detail factors affecting selection of location for the industrial plant. **[10]**

OR

Q2) a) Write short note on- **[9]**

- i) Line Layout
- ii) Travel chart
- iii) Functional Layout

b) Define material handling. List Material Handling equipment and describe any of them. **[8]**

Q3) a) Explain in brief various functions of production planning and control. **[8]**

b) Explain Moving Average Method for Demand forecasting. What is Line of balancing? Explain Assembly line balancing in detail. **[10]**

OR

Q4) a) What is Master Production Schedule (MPS)? Discuss objective and importance of MPS. **[9]**

P.T.O.

- b) Historical data on the sale of Laxmi Pumps for the last 12 years is given below. By the method of three yearly moving average establish the trend values and forecast demand for 13th year. If actual demand for 13th year is 520 nos. What shall be the forecast for 14th year? [9]

Year	1	2	3	4	5	6	7	8	9	10	11	12
Sales No.	332	344	328	336	370	408	420	432	428	450	498	540

- Q5) a)** Define Enterprise resource planning. Explain functions of Enterprise resource Planning system. [8]
b) What do you understand by term Cost Reduction & Value Improvement. Explain with example. What is Sampling Method. In detailed Acceptance Sampling and Inspection. [10]

OR

- Q6) a)** What is Inventory Control and Objectives of Inventory Control. [9]
b) Maxell manufacture has to supply his customers 3600 units of his products per year. Shortage is not permitted. Inventory carrying cost amounts Rs 1.2 per unit per annum. The setup per run is Rs. 80 Find. [9]
 i) Economic order quantity (EOQ)
 ii) Optimum number of orders per annum
 iii) Average annual inventory cost (minimum)

- Q7) a)** Explain the terms Rapid Upper Limb Assessment (RULA) and Rapid Entire Body Assessment (REBA). [6]
b) What is Performance Appraisal? Explain the types of Performance Appraisal methods. [6]
c) Explain basic principles of Ergonomics. [5]

OR

- Q8) a)** What is Merit Rating? Enlist and explain the methods of Merit Rating in Performance appraisal. [9]
b) What is value Engineering. Write Principles and benefits of value engineering in industry. [8]

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